Introduction To Machine Learning

KJR

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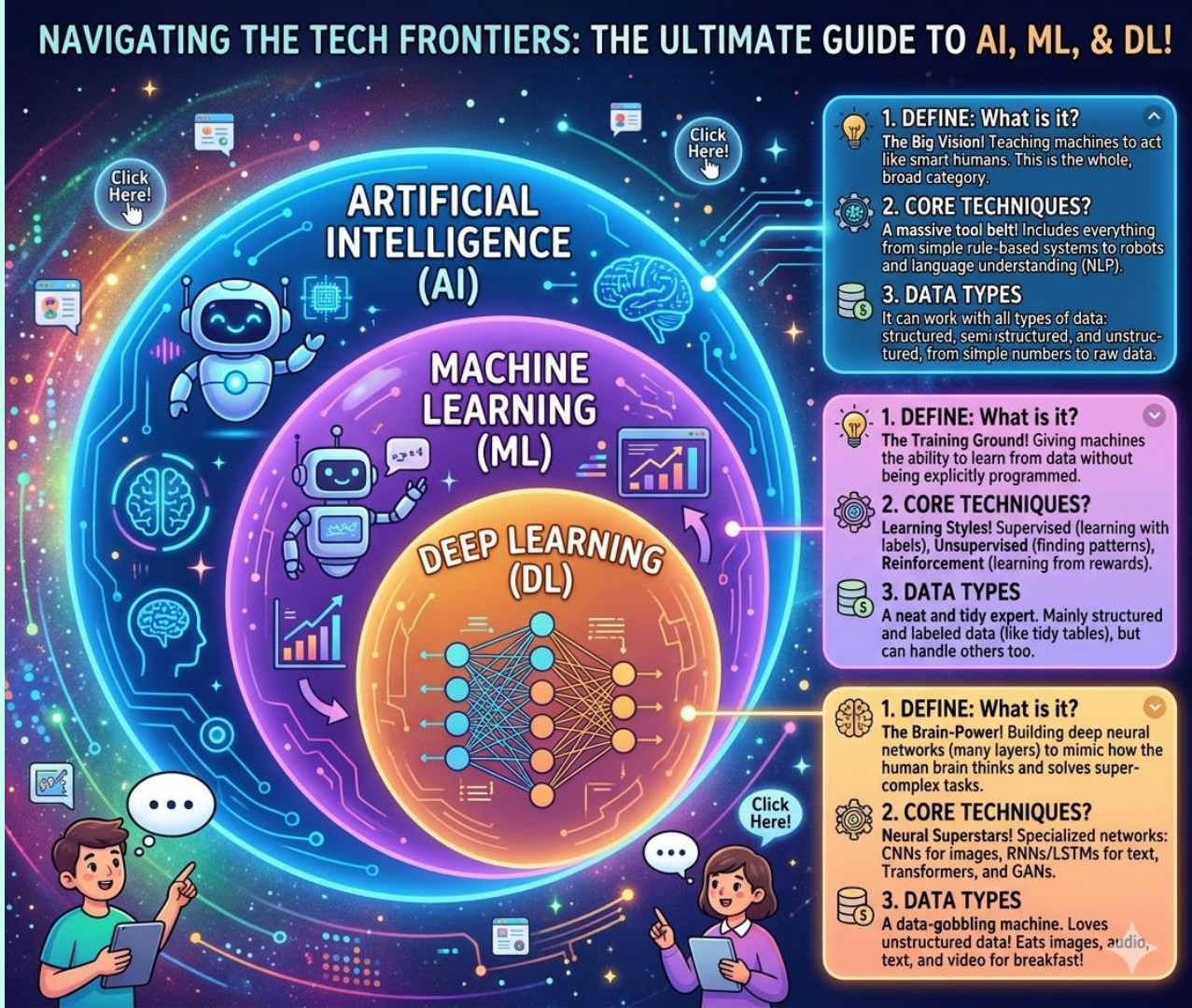
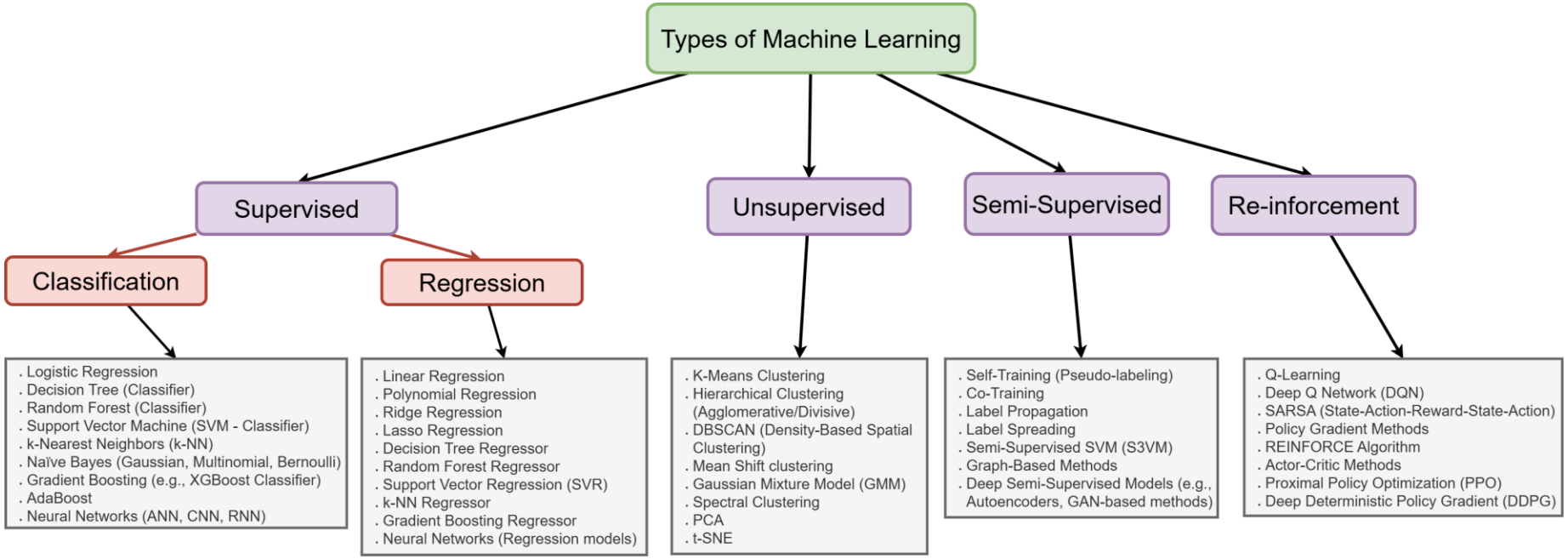
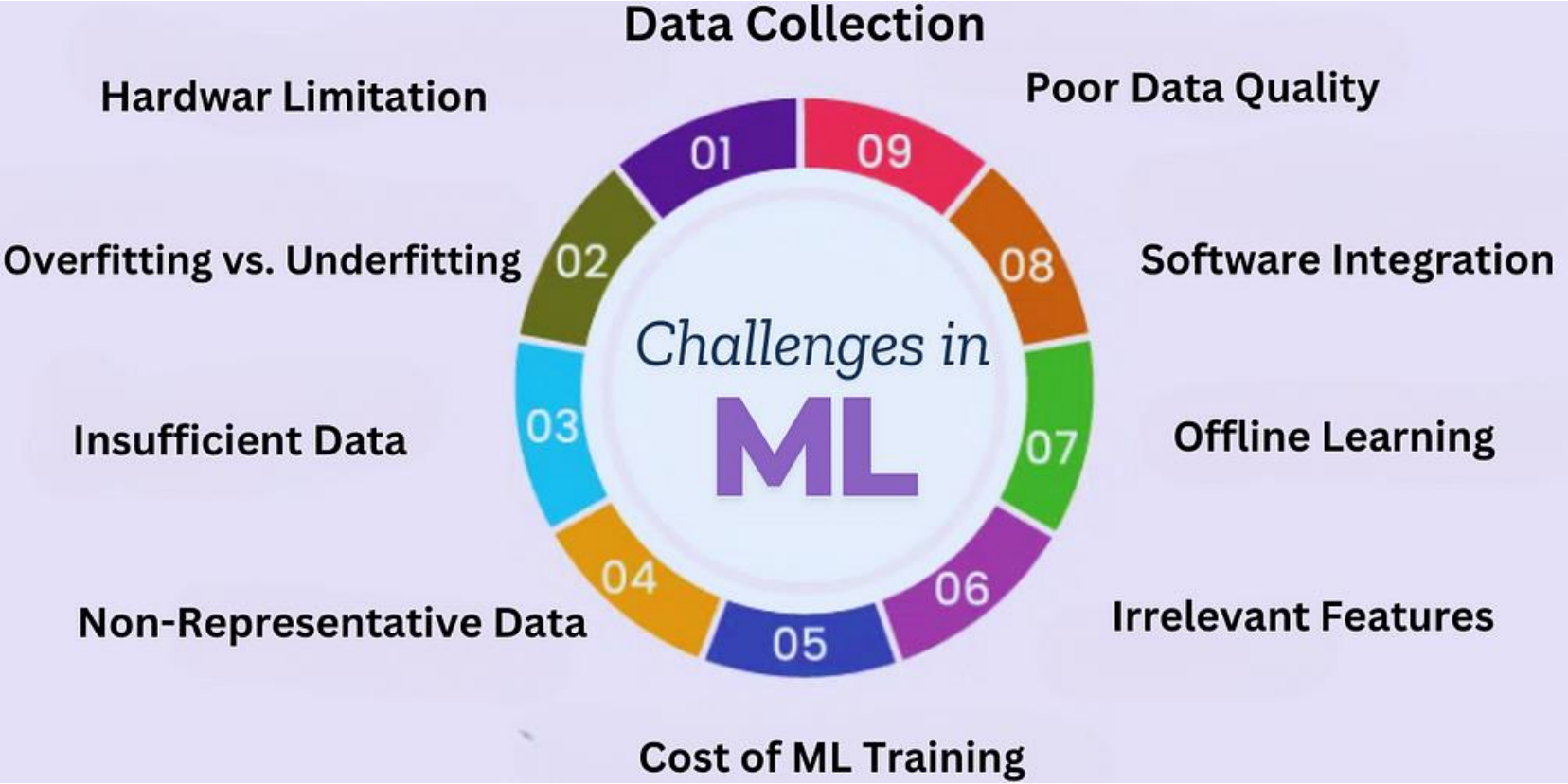
Difference Among
Artificial Intelligence (AI)
Machine Learning (ML)
Deep Learning (DL)



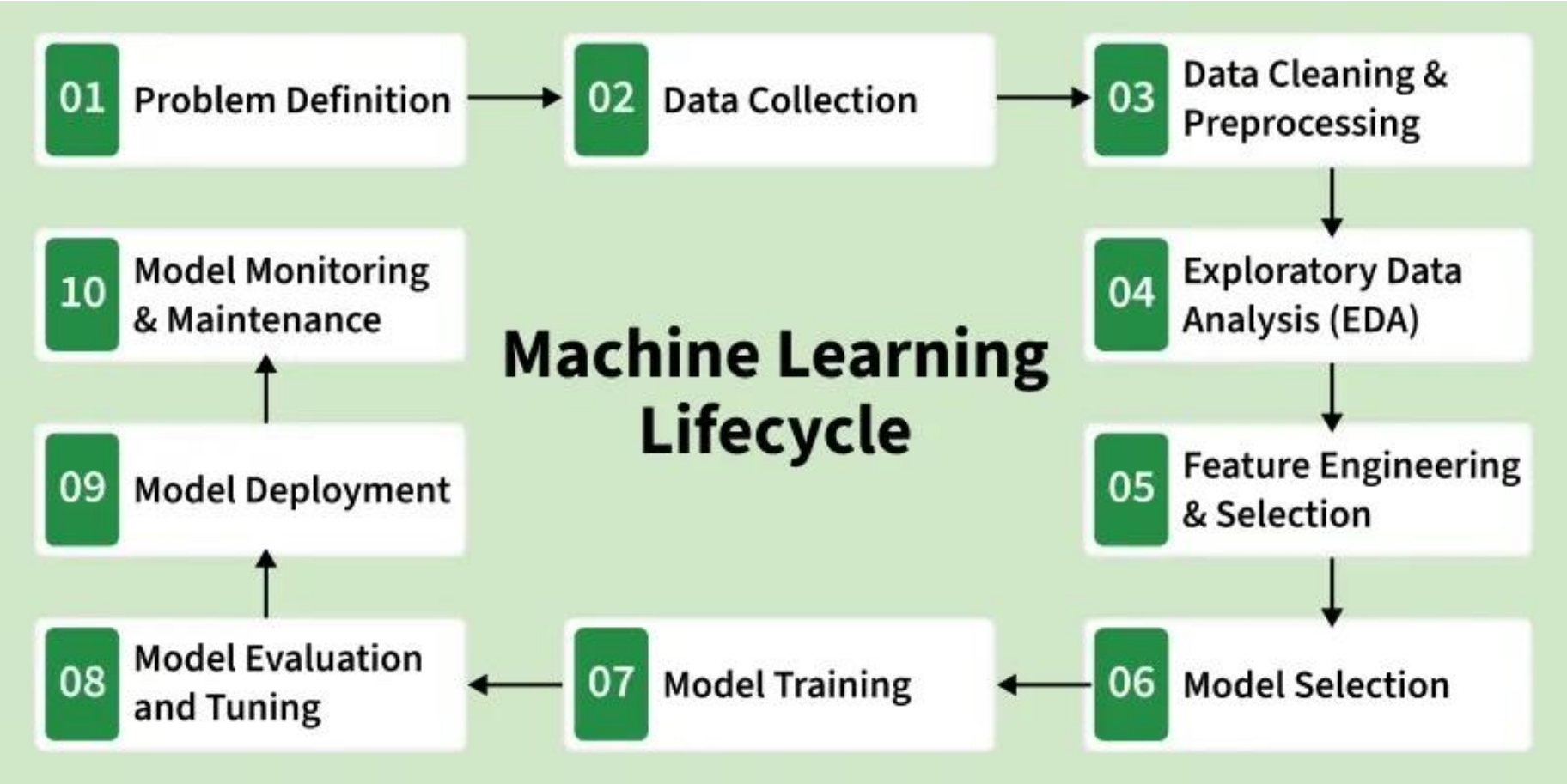

Types of Machine Learning
Supervised
Unsupervised
Semi-Supervised
Re-inforcement
Classification
Regression
. Logistic Regression
. Decision Tree (Classifier)
. Random Forest (Classifier)
. Support Vector Machine (SVM - Classifier)
. k-Nearest Neighbors (k-NN)
. Naïve Bayes (Gaussian, Multinomial, Bernoulli)
. Gradient Boosting (e.g., XGBoost Classifier)
. AdaBoost
. Neural Networks (ANN, CNN, RNN)
. Linear Regression
. Polynomial Regression
. Ridge Regression
. Lasso Regression
. Decision Tree Regressor
. Random Forest Regressor
. Support Vector Regression (SVR)
. k-NN Regressor
. Gradient Boosting Regressor
. Neural Networks (Regression models)
. K-Means Clustering
. Hierarchical Clustering (Agglomerative/Divisive)
. DBSCAN (Density-Based Spatial Clustering)
. Mean Shift clustering
. Gaussian Mixture Model (GMM)
. Spectral Clustering
. PCA
. t-SNE
. Self-Training (Pseudo-labeling)
. Co-Training
. Label Propagation
. Label Spreading
. Semi-Supervised SVM (S3VM)
. Graph-Based Methods
. Deep Semi-Supervised Models (e.g., Autoencoders, GAN-based methods)
. Q-Learning
. Deep Q Network (DQN)
. SARSA (State-Action-Reward-State-Action)
. Policy Gradient Methods
. REINFORCE Algorithm
. Actor-Critic Methods
. Proximal Policy Optimization (PPO)
. Deep Deterministic Policy Gradient (DDPG)

Challenges of ML



Applications of ML

Machine Learning Lifecycle



Different Job Roles

Data Engineer

1. Definition:

What does a Data Engineer do?

THEY BUILD THE PIPES!

They design the super-duper water slides for all our data.
They make sure data can flow easily from where it's kept to where the other smart folks can use it. They keep the data clean and organized!

2. Skills Required

- Being a builder of data slides!
- Working with big, complex sets of data.
- Good at organizing things.
- Understanding how databases work.

3. Tools:

SQL, Hadoop, Cloud Services

Data Analyst

1. Definition:

What does a Data Analyst do?

THEY TELL DATA STORIES!

They look closely at data to find cool secrets and answers.
For example: "Why do we sell more cookies on Mondays?"
They turn boring numbers into exciting picture reports to help make good decisions.

2. Skills:

- Spotting clues and patterns in data.
- Creating cool charts and graphs.
- Being good at simple math and logic.
- Communicating clearly so everyone understands the data story.

3. Tools:

SQL, X, +, ++++, ▥

Data Scientist

1. Definition:

What does a Data Scientist do?

THEY MAKE PREDICTIONS & RECIPES!

They build special number recipes, like spells, that can guess what will happen next. For example: "Which movie will you like the most?" or "How many ice creams will we sell?" They use fancy math to find the biggest insights.

2. Skills:

- Making smart number guesses with Machine Learning.
- Understanding big, tricky math (Statistics & Probability).
- Wrangling messy data into nice sets.
- Figuring out what info is most important (Feature Eng).

3. Tools:

Python, R, Neural Network

ML Engineer

1. Definition:

What does an ML Engineer do?

THEY MAKE MACHINES SMART!

They take the recipes and models from Data Scientists and make them work in the real world for millions of people. For example, the smart software that gives you movie recommendations. They ensure the AI is fast and reliable!

2. Skills:

- Creating smart 'Machine Learning' systems.
- Knowing how to code with Python or Java.
- Working with powerful computer brains (Big Data Tools like Spark).
- Building really fast, strong computer systems.

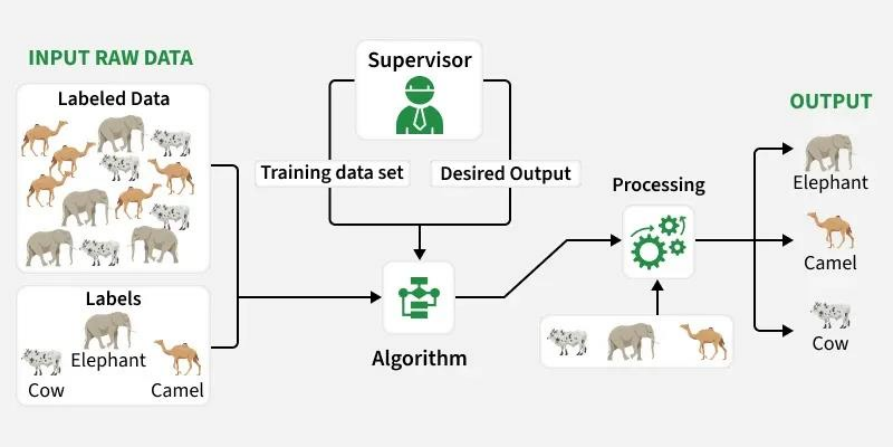
3. Tools:

TensorFlow, PyTorch, Spark like Hadoop

Types of Learning

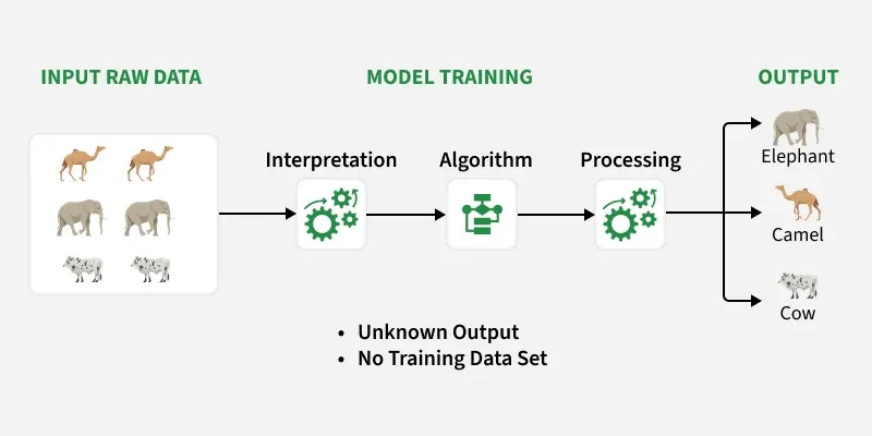
1. Supervised Machine Learning

Supervised learning is defined as when a model gets trained on a "Labeled Dataset". Labelled datasets have both input and output parameters. In Supervised Learning algorithms learn to map points between inputs and correct outputs. It has both training and validation datasets labelled.



2. Unsupervised Machine Learning

Unsupervised Learning works with unlabeled data, meaning there are no predefined outputs. The algorithm finds hidden patterns, groups or relationships within the data on its own. It's mainly used for clustering, dimensionality reduction and data visualization.



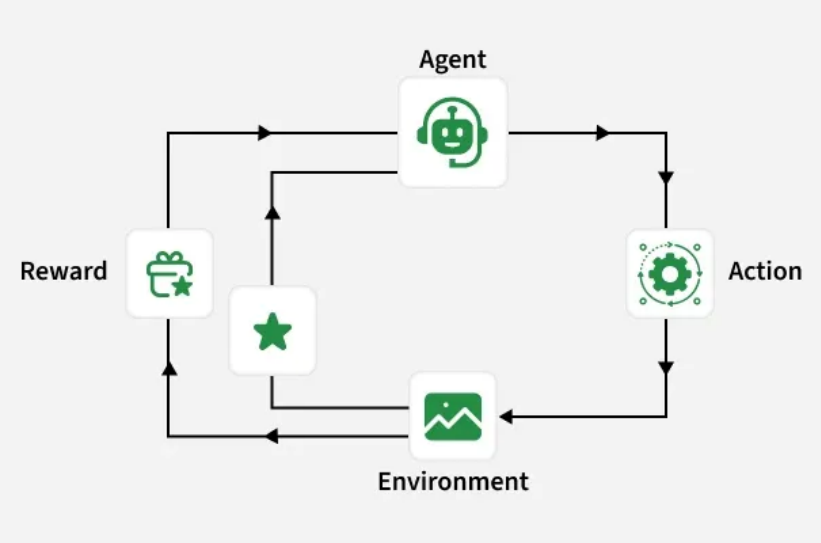
Types of Learning

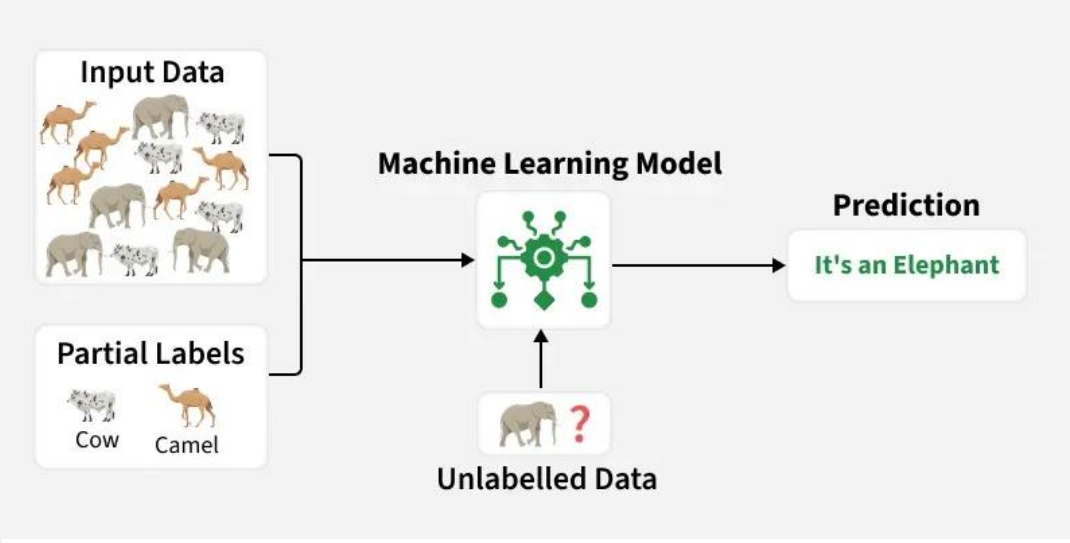
3. Reinforcement Learning

Reinforcement learning trains an agent to make a sequence of decisions through trial and error. The agent interacts with the environment, receives feedback in the form of rewards or penalties and learns optimal actions over time.

4. Semi-Supervised Learning: Supervised + Unsupervised Learning

Semi-Supervised learning Semi-Supervised Learning combines both Supervised and Unsupervised approaches. It uses a small set of labeled data and a large set of unlabeled data for training useful when labeling is costly or time-consuming.





Overview comparison between these methods

Category	Supervised	Unsupervised	Semi-supervised	Reinforcement
Input data	All data is labelled	All data is unlabelled	Partially labelled	No predefined data
Training?	External supervision	No supervision	(External supervision)	No supervision
Use	Calculate outcomes	Discover underlying patterns	Improve learning performance	Learn a series of outcomes
Computational complexity	Simple	Complex	Depends	Complex
Accuracy	Higher	Lesser	Lesser	Good for trial/error situations

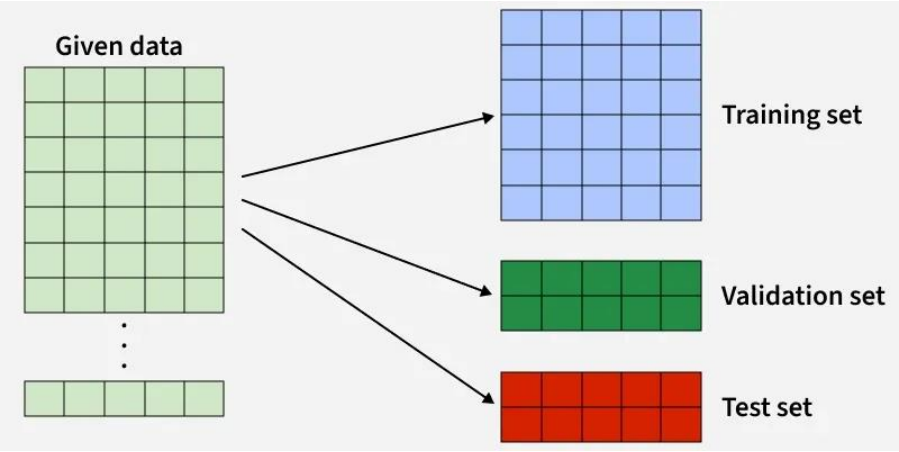
❑ Training vs Testing vs Validation Sets

Training, validation and testing sets are three essential components in building reliable machine learning models. The training set teaches the model patterns, the validation set helps fine-tune hyperparameters and prevent overfitting and the testing set evaluates how well the model performs on completely unseen data.

•**Training:** used to learn patterns

•**Validation:** used to tune and optimize the model

•**Testing:** used to measure final performance and generalization



❑ Cross Validation in Machine Learning

Cross-validation is a technique used to check how well a machine learning model performs on unseen data while preventing overfitting. It works by:

•Splitting the dataset into several parts.

•Training the model on some parts and testing it on the remaining part.

•Repeating this resampling process multiple times by choosing different parts of the dataset.

•Averaging the results from each validation step to get the final performance.

Read: Bangla Book for clear explanation

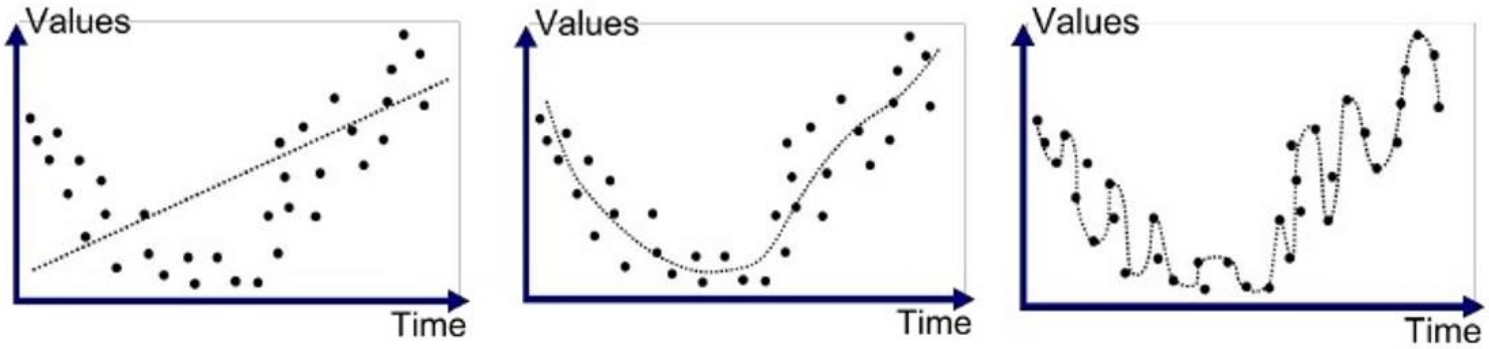
❑ Overfitting/underfitting

What is Overfitting?

Overfitting happens when your model learns the training data *too well*. It memorizes the data instead of generalizing patterns. As a result, it performs brilliantly on training data but poorly on test or real-world data.

What is Underfitting?

Underfitting is the opposite. It occurs when your model is too simple to capture the underlying patterns in the data. This often results in poor performance on both training and testing sets.



Underfitted Good Fit/Robust Overfitted

NEXT:

- Install Jupyter Notebook(anaconda/vscode/colab)
- Python basics required for ML

See You in the Next Lecture.....